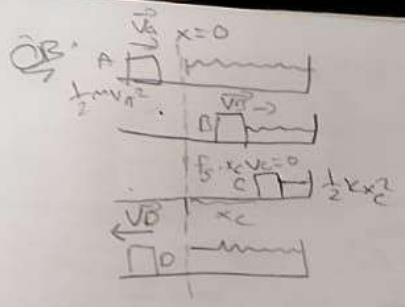
ÖR.



$m = 0{,}80$ kg

$V_A = 1{,}2$ m/s

$k = 50$ N/m

Sürtünme yokken

a) Yayın maksimum sıkışmasını

b) $\mu_k = 0{,}50$ iken yayın maksimum sıkışmasını bulunuz.

a) $\frac{1}{2} m V_A^2 = \frac{1}{2} k x_c^2$

$0{,}80 \cdot (1{,}2)^2 = 50 \cdot x_c^2$

$\boxed{x_c = 0{,}152\,m}$

b) $\frac{1}{2} m V_A^2 - f_k \cdot x_{c_x} = \frac{1}{2} k x_{c_2}^2$

$\frac{1}{2} m V_A^2 - \mu_k m g \cdot x_{c_x} = \frac{1}{2} k x_{c_2}^2$

$\frac{1}{2} 0{,}80 (1{,}2)^2 - 0{,}50 \cdot 0{,}80 \cdot 9{,}81\, x_{c_x} = \frac{1}{2} \cdot 50\, x_{c_2}^2$

$0{,}576 - 3{,}924\, x_{c_{x_2}} = 25 x_{c_2}^2$

$25 \cdot x_{c_2}^2 + 3{,}924\, x_{c_2} - 0{,}576 = 0$ $\boxed{x_{c_2} = 0{,}09\,m}$

DOĞRUSAL MOMENTUM VE ÇARPIŞMALAR

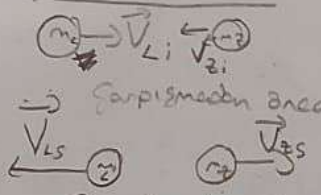
$\vec{P} = m \cdot \vec{V}$

$\vec{F} = \frac{d\vec{P}}{dt} = \frac{d}{dt}(m \cdot \vec{V})$

$\vec{F} = m \cdot \frac{d\vec{V}}{dt}$

$\vec{F} = m\vec{a}$

Esnek Çarpışma



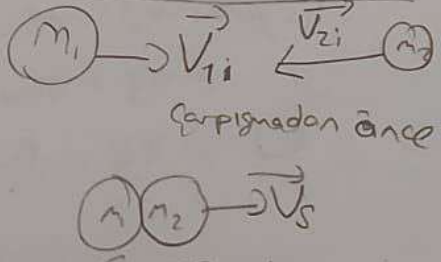
$\vec{P_i} = \vec{P_s}$

$m_1 \vec{V_{1i}} + m_2 \vec{V_{2i}} = m_1 \vec{V_{1s}} + m_2 \vec{V_{2s}}$

$E_i = E_s$

$\frac{1}{2} m_1 V_{1i}^2 + \frac{1}{2} m_2 V_{2i}^2 = \frac{1}{2} m_1 V_{1s}^2 + \frac{1}{2} m_2 V_{2s}^2$

Esnek olmayan çarpışma

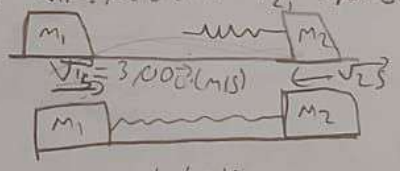


Enerji korunumu yoktur.

$\vec{P_i} = \vec{P_s}$

$m_1 \vec{V_{1i}} + m_2 \vec{V_{2i}} = (m_1 + m_2) \vec{V_s}$

ÖR. $\vec{V_{1i}} = 4{,}00\,\vec{i}$ (m/s) $\vec{V_{2i}} = -2{,}50\,\vec{i}$ (m/s)



$m_1 = 1{,}60$ kg

$m_2 = 2{,}10$ kg

$k = 600$ N/m

a) m_1 kütlesi sağa doğru 3,00 m/s hızla hareket ettiği anda 2. bloğun hızını bulunuz.

b) yaydaki sıkışma miktarını bulunuz.

a) $\vec{P_i} = \vec{P_s}$

$m_1 \vec{V_{1i}} + m_2 \vec{V_{2i}} = m_1 \vec{V_{1s}} + m_2 \vec{V_{2s}}$

$1{,}60 \cdot 4{,}00 + 2{,}10(-2{,}50) = 1{,}60 \cdot 3{,}00 + 2{,}10 V_{2s}$

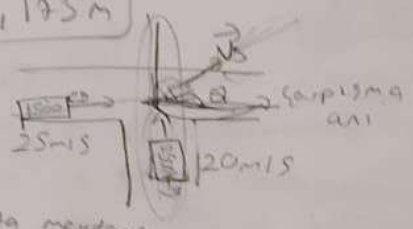
$\boxed{V_{2s} = -1{,}74\,m/s}$

b) $\frac{1}{2} m_1 V_i^2 + \frac{1}{2} m_2 V_{2i}^2 = \frac{1}{2} m_1 V_s^2 + \frac{1}{2} m_2 V_{2s}^2 + \frac{1}{2} k x^2$

$1,60 \cdot (4,00)^2 + 2,10\,(2,50)^2 = 1,60\,(3,00)^2 + 2,10\,(1,34)^2 + 600\,x^2$

$\boxed{x = 0,175\,m}$

Ör:



Karşıda meydana gelen bu çarpışma için ekrazin hızının büyüklüğü ve yönünü bulunuz

$V_S = ?$

$\theta = ?$

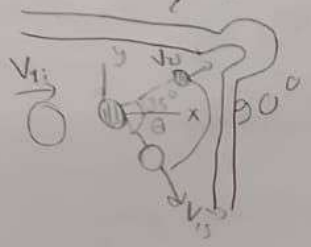
$$\frac{25 \cdot 1500 = 4000 \cdot V_S \cdot \cos\theta}{20 \cdot 1500 = 4000\, V_S \cdot \sin\theta} = \frac{3}{4} = \cot\theta \quad \boxed{\theta = 53,1^\circ}$$

$25 \cdot 1500 = 4000 \cdot V_S \cdot \cos 53,1^\circ$ //

$\boxed{V_S = 15,6\,m/s}$ //

(SINAVDA ÇIKACAK)

Ör:



Çarpışmanın esnek olduğu varsayılırsa gelen top hangi açı ile saçılır.

$$\frac{1}{2} m_1 V_{1i}^2 = \frac{1}{2} m_1 V_{1S}^2 + \frac{1}{2} m_2 V_{2S}^2$$

$$\boxed{V_{1i}^2 = V_{1S}^2 + V_{2S}^2 \quad (1)}$$

$$m_1 \vec{V_{1i}} = m_1 \vec{V_{1S}} + m_2 \vec{V_{2S}}$$

$$\boxed{\vec{V_{1i}} = \vec{V_{1S}} + \vec{V_{2S}} \quad (2)}$$

$$(\vec{V_{1i}})^2 = (\vec{V_{1S}} + \vec{V_{2S}})^2$$

$$\boxed{V_{1i}^2 = V_{1S}^2 + 2 \cdot \vec{V_{1S}} \cdot \vec{V_{2S}} + V_{2S}^2 \quad (3)}$$

$V_{1S} \cdot V_{2S} \cdot \cos\theta$

Bilinmeyenleri azaltmak için (3). denklemden (1). denklemi çıkarırım.

$$V_{1i}^2 = V_{1S}^2 + V_{2S}^2 + 2 V_{1S} \cdot V_{2S} \cdot \cos\theta \quad (3)$$

$$-\ V_{1i}^2 = V_{1S}^2 + V_{2S}^2 \quad (1)$$

$$0 = 2 \cdot V_{1S} \cdot V_{2S} \cdot \cos\theta \rightarrow 90^\circ$$

$\theta = 55^\circ$

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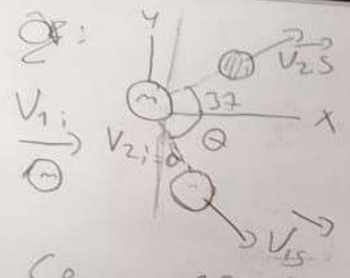
Ör:



Gelen protonun ilk hızı $3,5.10^5$ m/s dir ve ikinci protonla esnek çarpışma yapmaktadır. İki protonun son hızları ve saçılma açısını bulunuz

Cevap: 53,0°

$V_{1s} = 2,80.10^5$ m/s

$V_{2s} = 2,10.10^5$ m/s

$$\frac{1}{2}\cancel{m}V_{1i}^2 = \frac{1}{2}\cancel{m}V_{2s}^2 + \frac{1}{2}\cancel{m}V_{1s}^2$$

$$\boxed{V_{1i}^2 = V_{2s}^2 + V_{1s}^2}$$ 1. denklem

2. denklem $$\boxed{\cancel{m}.\vec{V_{1i}} = \cancel{m}.\vec{V_{1s}} + \cancel{m}\vec{V_{2s}}}$$

$$\vec{V_{1i}} = \vec{V_{1s}} + \vec{V_{2s}}$$

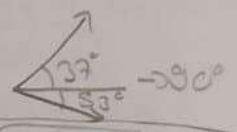
$V_{1s}.V_{2s}.\cos\theta$

$$\boxed{V_{1i}^2 = V_{1s}^2 + V_{2s}^2 + 2.\vec{V_{1s}}.\vec{V_{2s}}}$$ 3. denklem

$$V_{1i}^2 = V_{1s}^2 + V_{2s}^2 + V_{1s}.V_{2s}.\cos\theta$$

bilinmeyenleri azaltmak için 3. denklemden 1. denklem çıkarılır.

$$V_{1i}^2 = V_{1s}^2 + V_{2s}^2 + V_{1s}.V_{2s}.\cos\theta$$
$$- \quad V_{1i}^2 = V_{2s}^2 + V_{1s}^2$$



$$V_{1s}.V_{2s}.\cos\theta = 0 \qquad \cos\theta = 0 \qquad \theta = 90$$

$\boxed{\theta = 53°}$ saçılma açısı

$$-V_{1s}.\sin 53 + V_{2s}.\sin 37 = 0$$

$$V_{2}\sin 37 = V_{1s}\sin 53$$

$$V_{2s}.0,60 = V_{1s}\,0,79$$

k = oranti sabiti

$$\frac{V_{1s}}{V_{2s}} = \frac{0,60}{0,79} = \text{oranti}$$

$$\frac{V_{2s}}{V_{1s}} \simeq \frac{0,80}{0,60} \simeq \frac{4}{3} = \text{oranti}$$

$$(3,5.10^5)^2 = (3k)^2 + (4k)^2 = (3,5.10^5)^2 = 25k^2 = (5k)^2$$

$$(3,5.10^5)^2 = (5k)^2$$

$$\boxed{(0,7.10^5) = k}$$

$$V_{2s} = (0,7.10^5).4 = \boxed{2,8.10^5 = V_{2s}}$$

$$V_{1s} = (0,7.10^5).3 = \boxed{2,1.10^5 = V_{1s}}$$